

CC 352

Computer Graphics

Lecture 11

Three dimensional geometric
transformation



Basic 3D Transformations

- Position (translation)
- Size (scaling)
- Orientation (rotation)
- Shapes (shear)

3D Translation

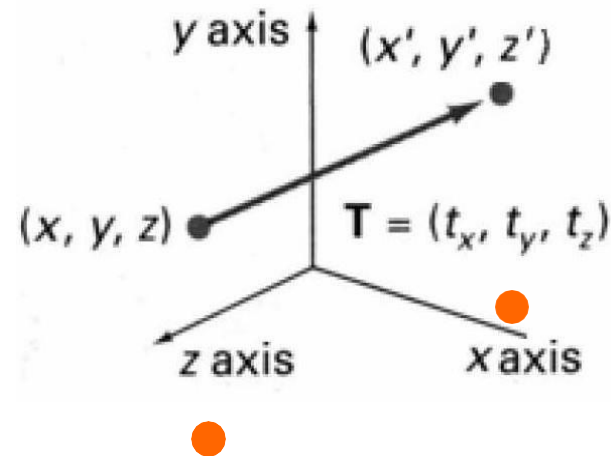
- Moving of object is called translation.
- In 3 dimensional homogeneous coordinate representation, a point is transformed from position

$$P = (x, y, z) \text{ to } P' = (x', y', z')$$

- This can be written as:-

Using $P' = T \cdot P$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



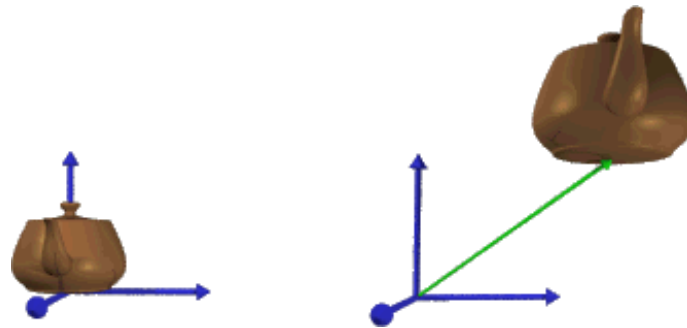
3D Translation

- The matrix representation is equivalent to the three equation.

$$x' = x + t_x,$$

$$y' = y + t_y,$$

$z' = z + t_z$ where parameter t_x, t_y, t_z are specifying translation distance for the coordinate direction x, y, z are assigned any real value.





Example

Given a 3D object with coordinate points A(0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0).

Apply the translation with the distance 1 towards X axis, 1 towards Y axis and 2 towards Z axis and obtain the new coordinates of the object.

For Coordinates A(0, 3, 1)

$$X_{\text{new}} = X_{\text{old}} + T_x = 0 + 1 = 1$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 3 + 1 = 4$$

$$Z_{\text{new}} = Z_{\text{old}} + T_z = 1 + 2 = 3$$

For Coordinates B(3, 3, 2)

$$X_{\text{new}} = X_{\text{old}} + T_x = 3 + 1 = 4$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 3 + 1 = 4$$

$$Z_{\text{new}} = Z_{\text{old}} + T_z = 2 + 2 = 4$$

For Coordinates C(3, 0, 0)

$$X_{\text{new}} = X_{\text{old}} + T_x = 3 + 1 = 4$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 0 + 1 = 1$$

$$Z_{\text{new}} = Z_{\text{old}} + T_z = 0 + 2 = 2$$

For Coordinates D(0, 0, 0)

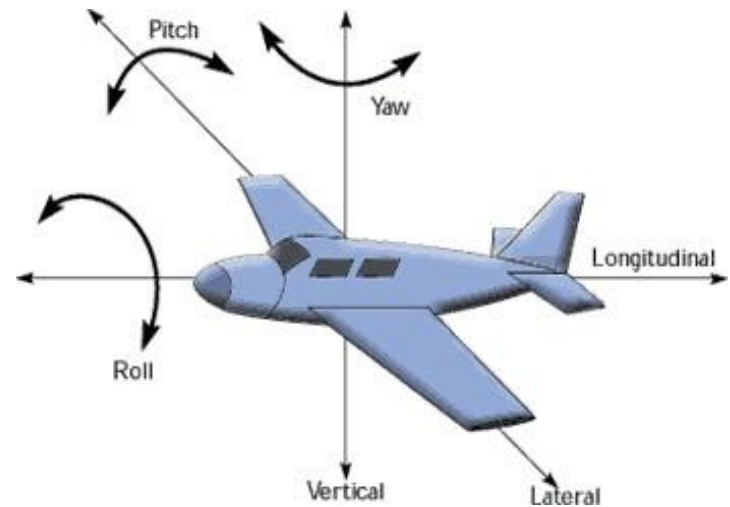
$$X_{\text{new}} = X_{\text{old}} + T_x = 0 + 1 = 1$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 0 + 1 = 1$$

$$Z_{\text{new}} = Z_{\text{old}} + T_z = 0 + 2 = 2$$

3D Geometric Rotations

- When we performed rotations in two dimensions we only had the choice of rotating about the z axis
- In the case of three dimensions we have more options
 - Rotate about x – pitch
 - Rotate about y – yaw
 - Rotate about z – roll





3D Geometric Rotations

- **Roll** is known as the rising or dipping of the airplane's wing. The movement is done about **Longitudinal axis**. The Ailerons controlling the roll are located on the trailing edge of both wings.
- **Pitch** refers to the movement of the airplane's nose either up or down. Its movement is done about **Lateral axis**. The elevator controlling the pitch is also located on the rear of the aircraft on the tail, along with the rudder.
- **Yaw** allows the airplane to move towards the left or right while in flight. The movement is done about a **ventricle axis**. The yaw is controlled by the rudder located in the rear of the aircraft on the tail.



Roll



Pitch

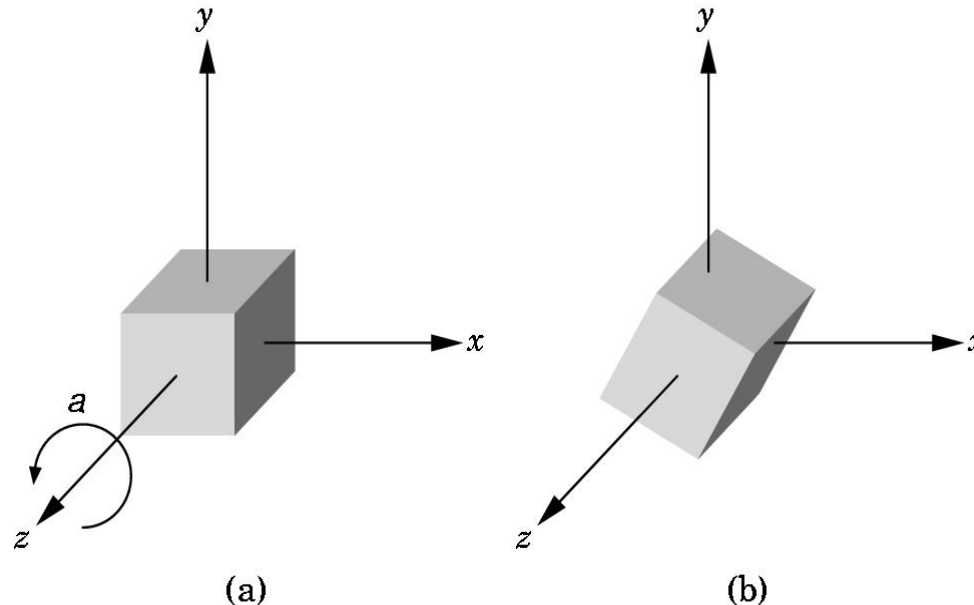


Yaw



3D Geometric Rotations

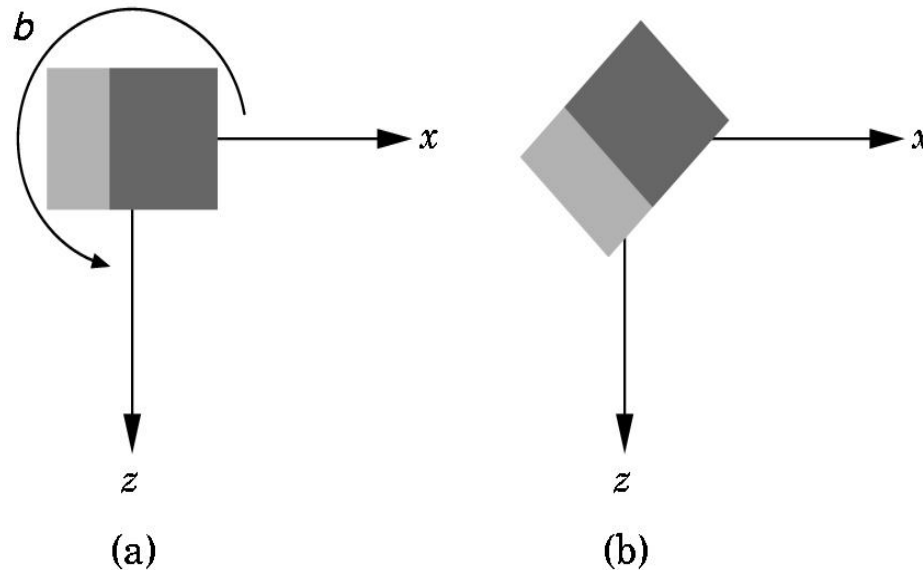
- Rotate about z – axis: The picture shows a z -axis rotation around the origin in a positive angle, (Anti-clockwise) as you look down the z -axis towards the origin. The angle is measured in the xy -plane from the x -axis, just as in 2D.





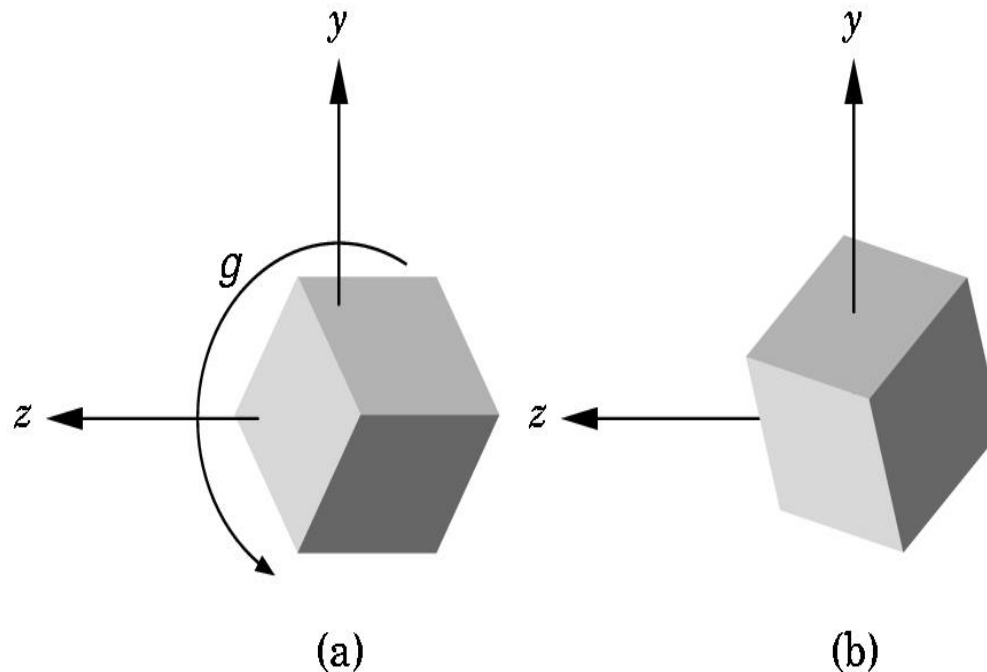
3D Geometric Rotations

- Rotate about y –axis: You are looking down the y -axis which is not shown. A positive (counter-clockwise) angle is shown.



3D Geometric Rotations

- Rotate about x – axis: You are looking down the x -axis which is not shown. A positive (counter-clockwise) angle is shown.



X-Axis Rotation

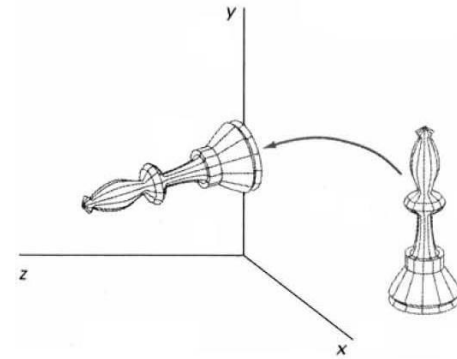
The equation for X-axis rotation

$$\mathbf{x}' = \mathbf{x}$$

$$\mathbf{y}' = \mathbf{y} \cos\theta - \mathbf{z} \sin\theta$$

$$\mathbf{z}' = \mathbf{y} \sin\theta + \mathbf{z} \cos\theta$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Y-Axis Rotation

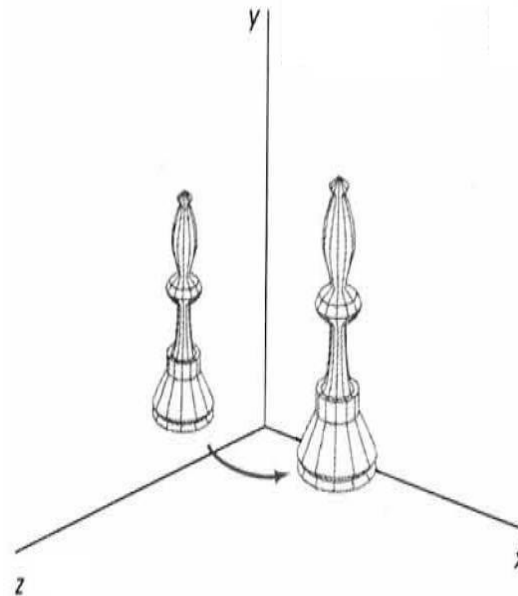
The equation for Y-axis rotation

$$x' = x \cos\theta + z \sin\theta$$

$$y' = y$$

$$z' = z \cos\theta - x \sin\theta$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$





Z-Axis Rotation

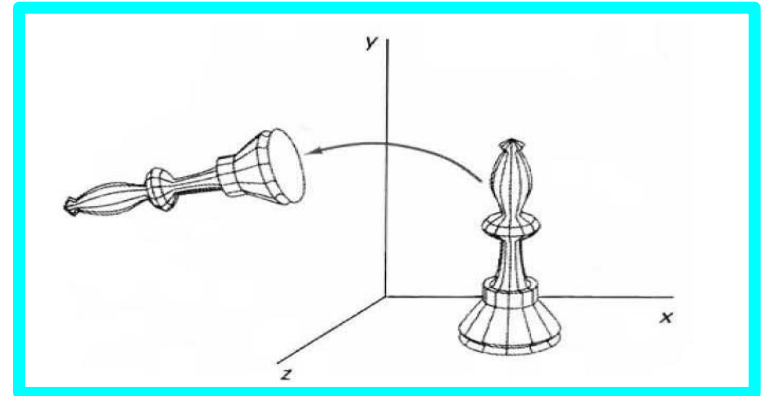
The equation for Z-axis rotation

$$x' = x \cos\theta - y \sin\theta$$

$$y' = x \sin\theta + y \cos\theta$$

$$z' = z$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



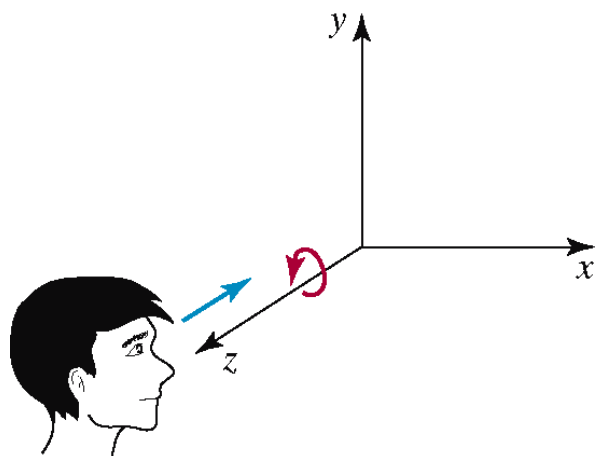


3D rotation

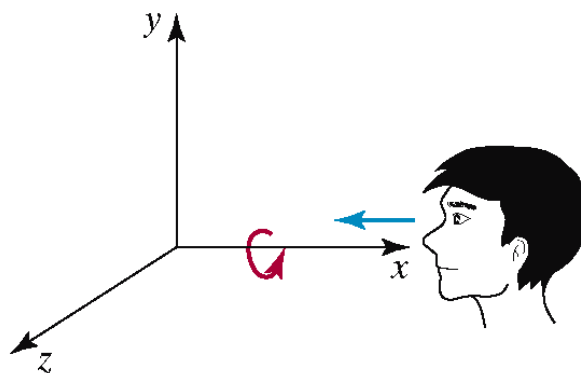
$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

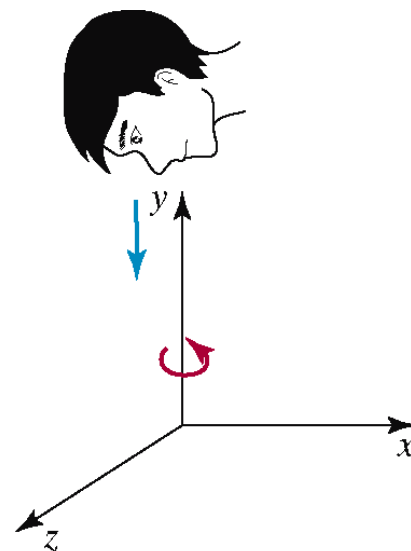
$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$



(a)



(b)



(c)



Example

Given a homogeneous point (1, 2, 3). Apply rotation 90° about X, Y and Z axis and find out the new coordinate points.

For X-Axis Rotation-

$$X_{\text{new}} = X_{\text{old}} = 1$$

$$Y_{\text{new}} = Y_{\text{old}} * \cos\theta - Z_{\text{old}} * \sin\theta = 2 * \cos 90^\circ - 3 * \sin 90^\circ = 2 * 0 - 3 * 1 = -3$$

$$Z_{\text{new}} = Y_{\text{old}} * \sin\theta + Z_{\text{old}} * \cos\theta = 2 * \sin 90^\circ + 3 * \cos 90^\circ = 2 * 1 + 3 * 0 = 2$$

For Y-Axis Rotation-

$$X_{\text{new}} = Z_{\text{old}} * \sin\theta + X_{\text{old}} * \cos\theta = 3 * \sin 90^\circ + 1 * \cos 90^\circ = 3 * 1 + 1 * 0 = 3$$

$$Y_{\text{new}} = Y_{\text{old}} = 2$$

$$Z_{\text{new}} = Y_{\text{old}} * \cos\theta - X_{\text{old}} * \sin\theta = 2 * \cos 90^\circ - 1 * \sin 90^\circ = 2 * 0 - 1 * 1 = -1$$

For Z-Axis Rotation-

$$X_{\text{new}} = X_{\text{old}} * \cos\theta - Y_{\text{old}} * \sin\theta = 1 * \cos 90^\circ - 2 * \sin 90^\circ = 1 * 0 - 2 * 1 = -2$$

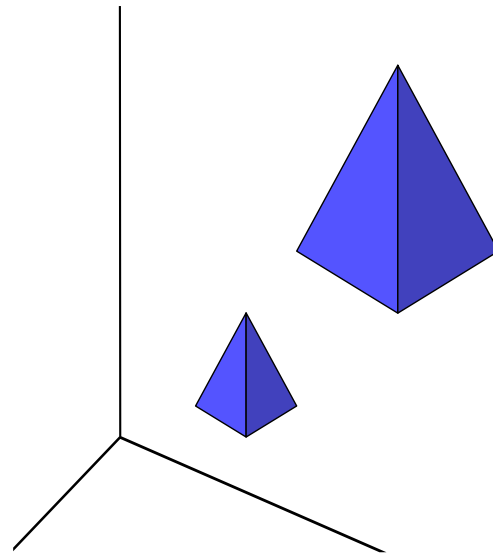
$$Y_{\text{new}} = X_{\text{old}} * \sin\theta + Y_{\text{old}} * \cos\theta = 1 * \sin 90^\circ + 2 * \cos 90^\circ = 1 * 1 + 2 * 0 = 1$$

$$Z_{\text{new}} = Z_{\text{old}} = 3$$

3D Scaling

- Changes the size of the object and repositions the object relative to the coordinate origin.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Scaling

- The equations for scaling

$$\mathbf{x}' = \mathbf{x} \cdot \mathbf{sx}$$

$$\mathbf{S}_{\mathbf{sx}, \mathbf{sy}, \mathbf{sz}} \quad \square \quad \mathbf{y}' = \mathbf{y} \cdot \mathbf{sy}$$

$$\mathbf{z}' = \mathbf{z} \cdot \mathbf{sz}$$





Example

Given a 3D object with coordinate points A(0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0).

Apply the scaling parameter 2 towards X axis, 3 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object.

For Coordinates A(0, 3, 3)

$$X_{\text{new}} = X_{\text{old}} * S_x = 0 * 2 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} * S_y = 3 * 3 = 9$$

$$Z_{\text{new}} = Z_{\text{old}} * S_z = 3 * 3 = 9$$

For Coordinates B(3, 3, 6)

$$X_{\text{new}} = X_{\text{old}} * S_x = 3 * 2 = 6$$

$$Y_{\text{new}} = Y_{\text{old}} * S_y = 3 * 3 = 9$$

$$Z_{\text{new}} = Z_{\text{old}} * S_z = 6 * 3 = 18$$

For Coordinates C(3, 0, 1)

$$X_{\text{new}} = X_{\text{old}} * S_x = 3 * 2 = 6$$

$$Y_{\text{new}} = Y_{\text{old}} * S_y = 0 * 3 = 0$$

$$Z_{\text{new}} = Z_{\text{old}} * S_z = 1 * 3 = 3$$

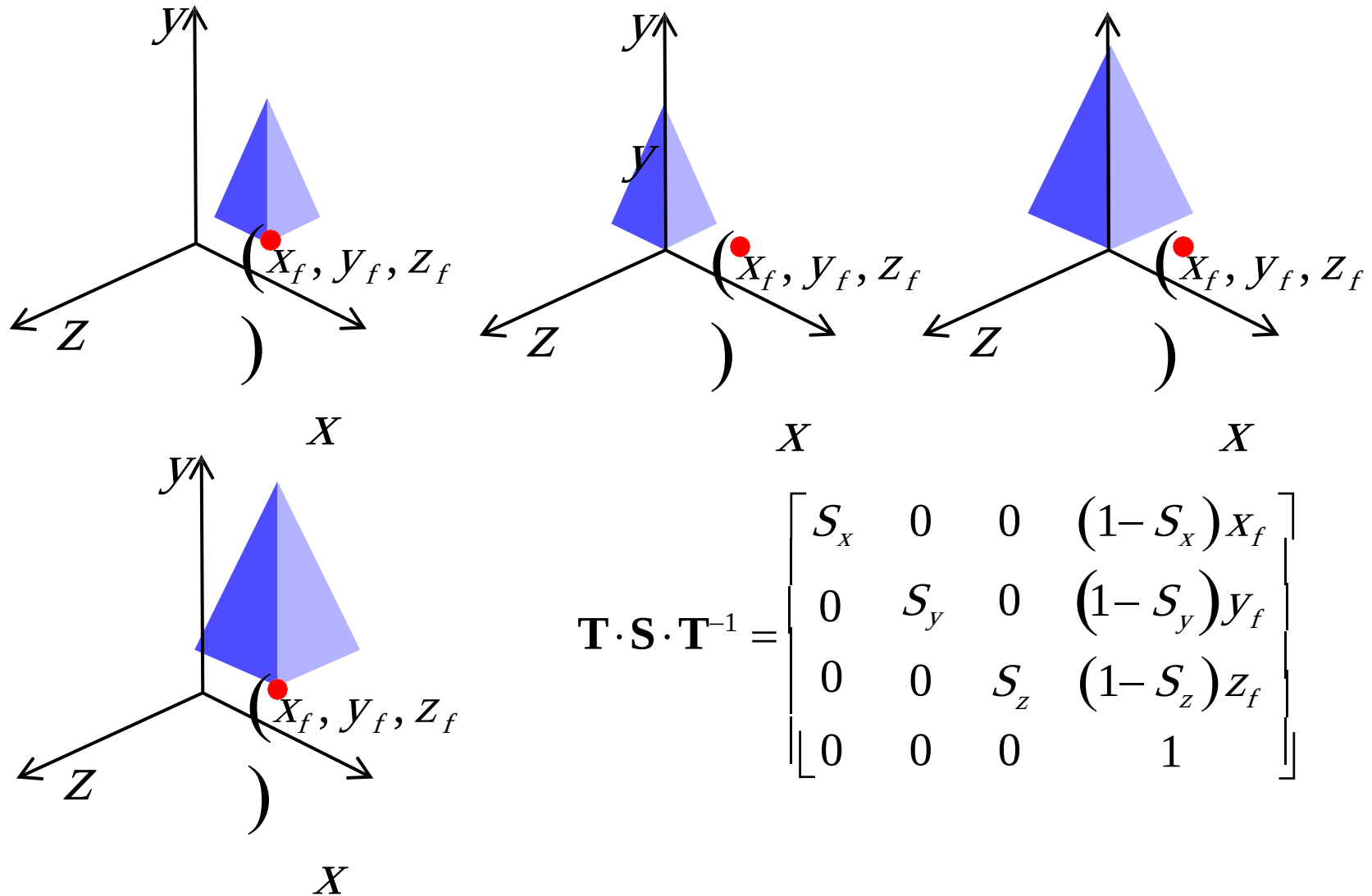
For Coordinates D(0, 0, 0)

$$X_{\text{new}} = X_{\text{old}} * S_x = 0 * 2 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} * S_y = 0 * 3 = 0$$

$$Z_{\text{new}} = Z_{\text{old}} * S_z = 0 * 3 = 0$$

Scaling with respect to a fixed point (not necessarily of object)



3D Reflection

- Reflection in computer graphics is used to emulate reflective objects like mirrors and shiny surfaces
- Reflection may be:
 - an x-axis, y-axis , z-axis. and also
 - in the planes xy-plane, yz-plane, and zx-plane.
- Reflection relative to a given Axis are
 - equivalent to 180 Degree rotations.



3D Reflection

➤ Reflection about x-axis:-

$$\mathbf{x}'=\mathbf{x} \quad \mathbf{y}'=-\mathbf{y} \quad \mathbf{z}'=-\mathbf{z}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



$$\begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}$$

Reflection about y-axis:-

$$\mathbf{y}'=\mathbf{y} \quad \mathbf{x}'=-\mathbf{x} \quad \mathbf{z}'=-\mathbf{z}$$



3D Reflection

- The matrix for reflection about y-axis:-

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- Reflection about z-axis:-

$$\mathbf{x}' = -\mathbf{x} \quad \mathbf{y}' = -\mathbf{y} \quad \mathbf{z}' = \mathbf{z}$$

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$





Example

- **Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XY plane and find out the new coordinates of the object.**

For Coordinates A(3, 4, 1)

$$X_{\text{new}} = X_{\text{old}} = 3$$

$$Y_{\text{new}} = Y_{\text{old}} = 4$$

$$Z_{\text{new}} = -Z_{\text{old}} = -1$$

For Coordinates B(6, 4, 2)

$$X_{\text{new}} = X_{\text{old}} = 6$$

$$Y_{\text{new}} = Y_{\text{old}} = 4$$

$$Z_{\text{new}} = -Z_{\text{old}} = -2$$

For Coordinates C(5, 6, 3)

$$X_{\text{new}} = X_{\text{old}} = 5$$

$$Y_{\text{new}} = Y_{\text{old}} = 6$$

$$Z_{\text{new}} = -Z_{\text{old}} = -3$$



Example

- Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XZ plane and find out the new coordinates of the object.

For Coordinates A(3, 4, 1)

$$X_{\text{new}} = X_{\text{old}} = 3$$

$$Y_{\text{new}} = -Y_{\text{old}} = -4$$

$$Z_{\text{new}} = Z_{\text{old}} = 1$$

For Coordinates B(6, 4, 2)

$$X_{\text{new}} = X_{\text{old}} = 6$$

$$Y_{\text{new}} = -Y_{\text{old}} = -4$$

$$Z_{\text{new}} = Z_{\text{old}} = 2$$

For Coordinates C(5, 6, 3)

$$X_{\text{new}} = X_{\text{old}} = 5$$

$$Y_{\text{new}} = -Y_{\text{old}} = -6$$

$$Z_{\text{new}} = Z_{\text{old}} = 3$$

3D Shearing

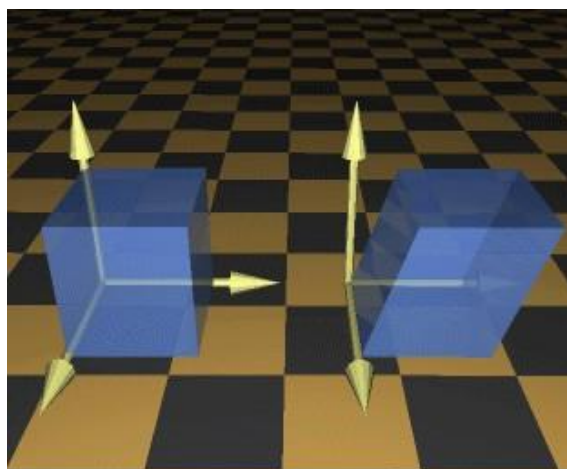
- **Modify object shapes**
- **Useful for perspective projections**
- When an object is viewed from different directions and at different distances, the appearance of the object will be different.
- Such view is called perspective view.
- Perspective projections mimic what the human eyes see.

e.g. draw a cube (3D) on a screen (2D) Alter the values for x and y by an amount proportional to the distance from z_{ref}



3D Shearing

- Matrix for 3d shearing
- Where a and b can be assigned any real value.



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Shearing

- In (y, z) w.r.t. x value

$$SH_{yz} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ sh_y & 1 & 0 & 0 \\ sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- In (z, x) w.r.t. y value

$$SH_{xz} = \begin{bmatrix} 1 & sh_x & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

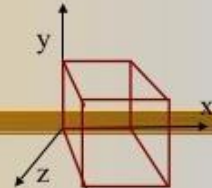
- In (x, y) w.r.t. z value

$$SH_{xy} = \begin{bmatrix} 1 & 0 & sh_x & 0 \\ 0 & 1 & sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Shearing



Shear along Z-axis


$$\begin{matrix} SH_{xy}(sh_x, sh_y) & * & P & = & P' \\ \Downarrow & & \Downarrow & & \Downarrow \\ \begin{bmatrix} 1 & 0 & sh_x & 0 \\ 0 & 1 & sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & * & \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} & = & \begin{bmatrix} x + z * sh_x \\ y + z * sh_y \\ z \\ 1 \end{bmatrix} \end{matrix}$$

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5) 3-D Shears

- Shears along planes:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ sh_y & 1 & 0 & 0 \\ sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & sh_x & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & sh_x & 0 \\ 0 & 1 & sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

y-z

x-z

x-y plane

- Shears parallel to an axis:

$$\begin{bmatrix} 1 & sh_{x_1} & sh_{x_2} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ sh_{y_1} & 1 & sh_{y_2} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ sh_{z_1} & sh_{z_2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

x

y

z axis



Example

- Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Shearing in X Axis-

For Coordinates A(0, 0, 0)

$$\begin{aligned}X_{\text{new}} &= X_{\text{old}} = 0 \\Y_{\text{new}} &= Y_{\text{old}} + Sh_y * X_{\text{old}} = 0 + 2 * 0 = 0 \\Z_{\text{new}} &= Z_{\text{old}} + Sh_z * X_{\text{old}} = 0 + 3 * 0 = 0\end{aligned}$$

For Coordinates B(1, 1, 2)

$$\begin{aligned}X_{\text{new}} &= X_{\text{old}} = 1 \\Y_{\text{new}} &= Y_{\text{old}} + Sh_y * X_{\text{old}} = 1 + 2 * 1 = 3 \\Z_{\text{new}} &= Z_{\text{old}} + Sh_z * X_{\text{old}} = 2 + 3 * 1 = 5\end{aligned}$$

For Coordinates C(1, 1, 3)

$$\begin{aligned}X_{\text{new}} &= X_{\text{old}} = 1 \\Y_{\text{new}} &= Y_{\text{old}} + Sh_y * X_{\text{old}} = 1 + 2 * 1 = 3 \\Z_{\text{new}} &= Z_{\text{old}} + Sh_z * X_{\text{old}} = 3 + 3 * 1 = 6\end{aligned}$$



Example

- Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Shearing in Y Axis-

For Coordinates A(0, 0, 0)

$$X_{\text{new}} = X_{\text{old}} + Sh_x * Y_{\text{old}} = 0 + 2 * 0 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} = 0$$

$$Z_{\text{new}} = Z_{\text{old}} + Sh_z * Y_{\text{old}} = 0 + 3 * 0 = 0$$

For Coordinates B(1, 1, 2)

$$X_{\text{new}} = X_{\text{old}} + Sh_x * Y_{\text{old}} = 1 + 2 * 1 = 3$$

$$Y_{\text{new}} = Y_{\text{old}} = 1$$

$$Z_{\text{new}} = Z_{\text{old}} + Sh_z * Y_{\text{old}} = 2 + 3 * 1 = 5$$

For Coordinates C(1, 1, 3)

$$X_{\text{new}} = X_{\text{old}} + Sh_x * Y_{\text{old}} = 1 + 2 * 1 = 3$$

$$Y_{\text{new}} = Y_{\text{old}} = 1$$

$$Z_{\text{new}} = Z_{\text{old}} + Sh_z * Y_{\text{old}} = 3 + 3 * 1 = 6$$



Example

- Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Shearing in Z Axis-

For Coordinates A(0, 0, 0)

$$X_{\text{new}} = X_{\text{old}} + Sh_x * Z_{\text{old}} = 0 + 2 * 0 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} + Sh_y * Z_{\text{old}} = 0 + 2 * 0 = 0$$

$$Z_{\text{new}} = Z_{\text{old}} = 0$$

For Coordinates B(1, 1, 2)

$$X_{\text{new}} = X_{\text{old}} + Sh_x * Z_{\text{old}} = 1 + 2 * 2 = 5$$

$$Y_{\text{new}} = Y_{\text{old}} + Sh_y * Z_{\text{old}} = 1 + 2 * 2 = 5$$

$$Z_{\text{new}} = Z_{\text{old}} = 2$$

For Coordinates C(1, 1, 3)

$$X_{\text{new}} = X_{\text{old}} + Sh_x * Z_{\text{old}} = 1 + 2 * 3 = 7$$

$$Y_{\text{new}} = Y_{\text{old}} + Sh_y * Z_{\text{old}} = 1 + 2 * 3 = 7$$

$$Z_{\text{new}} = Z_{\text{old}} = 3$$



Composite 3D transformation

A translation and rotation can be combined by multiplying the transformation matrices.

The translation terms do not affect the rotations.

E.g. $T(d_x, d_y, d_z) R_z(\theta) =$

1	0	0	d_x
0	1	0	d_y
0	0	1	d_z
0	0	0	1

*

$\cos(\theta)$	$-\sin(\theta)$	0	0
$\sin(\theta)$	$\cos(\theta)$	0	0
0	0	1	0
0	0	0	1

=

$\cos(\theta)$	$-\sin(\theta)$	0	d_x
$\sin(\theta)$	$\cos(\theta)$	0	d_y
0	0	1	d_z
0	0	0	1



Example

- The matrices are stacked in reversed order of operations from right-to-left
- Then matrix multiplications from right-to-left
- All transformations are with respect to the reference frame.

A point is translated in the y direction, rotated around the x-axis, translated in the z direction, and finally rotated around the y-axis. Find the composite transformation matrix.

$${}^R T_N = R_y(90) T(0,0,a) R_x(90) T(0,a,0)$$

← Order of transformations

← Multiplication sequence



Example

$\cos(90)$	0	$\sin(90)$	0
0	1	0	0
$-\sin(90)$	0	$\cos(90)$	0
0	0	0	1

1	0	0	0
0	1	0	0
0	0	1	a
0	0	0	1

1	0	0	0
0	$\cos(90)$	$-\sin(90)$	0
0	$\sin(90)$	$\cos(90)$	0
0	0	0	1

1	0	0	0
0	1	0	a
0	0	1	0
0	0	0	1

=

0	0	1	0
0	1	0	0
-1	0	0	0
0	0	0	1

1	0	0	0
0	1	0	0
0	0	1	a
0	0	0	1

1	0	0	0
0	0	-1	0
0	1	0	0
0	0	0	1

1	0	0	0
0	1	0	0
0	0	1	a
0	0	0	1



Example

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & a \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2a \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{array}{c} \begin{matrix} X' & Y' & Z' & \text{dis} \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 & 2a \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$